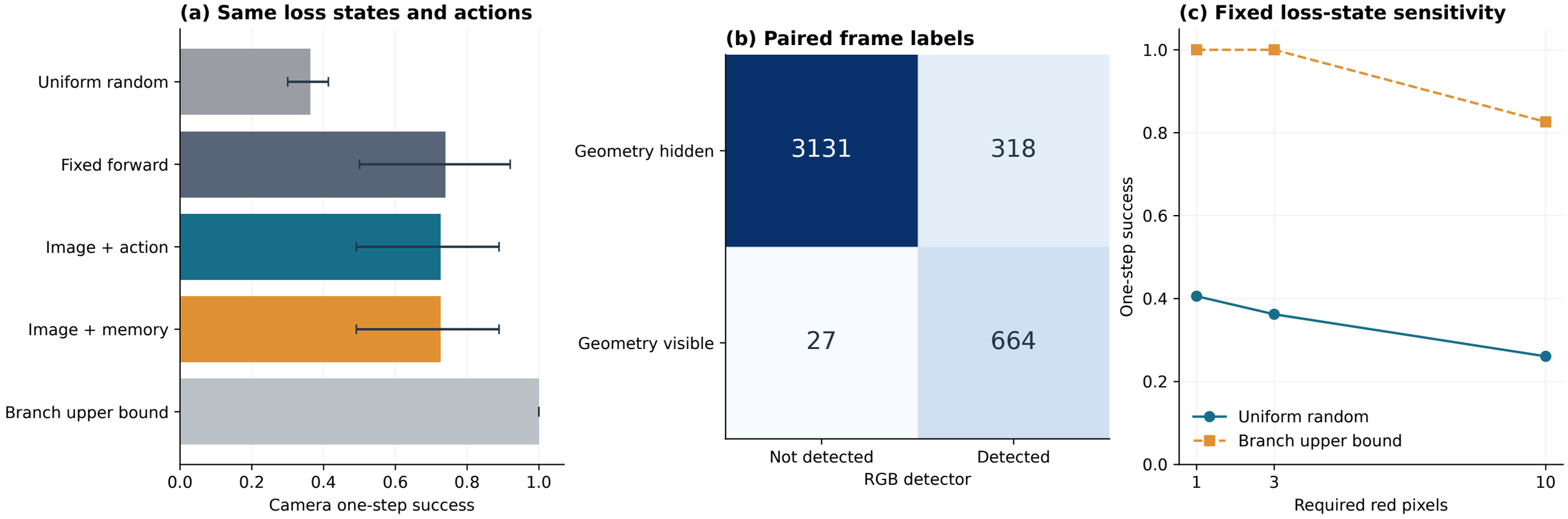

(a) Same loss states and actions
Uniform random
Fixed forward
Image + action
Image + memory
Branch upper bound
0.0
0.2
0.4
0.6
0.8
1.0
Camera one-step success
(b) Paired frame labels
Geometry hidden
Geometry visible
3131
318
27
664
Not detected
Detected
RGB detector
(c) Fixed loss-state sensitivity
One-step success
0.0
0.2
0.4
0.6
0.8
1.0
Uniform random
Branch upper bound
1
3
10
Required red pixels